

Name _____

Environments and Closures Practice In Class

DAY 2

In Racket, what order are the procedures A B C D E called?

A (1 point)

```
((lambda (fun1 fun2)
```

```
  (A 99)
```

```
  (fun1 99)
```

```
  (fun2 99))
```

B

```
(C 99))
```

B (1 point)

```
(let ([myFuncs (list A B)])
```

```
  ((cadr myFuncs)
```

```
    ((car myFuncs) 7)
```

```
    (C D)))
```

Draw E&C Diagrams for the following code:

A (4 points)

```
(let ([A (lambda (x) (display 'A))]
      [B (lambda (x y) (display 'B))]
      [C (lambda (x) (display 'C))]
      [D (lambda (x) (display 'D))])
  (let ([myFuncs (list A B)])
    ((cadr myFuncs)
     ((car myFuncs) 7)
     (C D))))
```

B (4 points) <- if you finish this and get it checked off, you can leave

```
(let ((make-count-down
      (lambda (init)
        (let [(num init)]
          (lambda (deduct)
            (set! num (- num deduct))
            num))))))
  (let ((cd1 (make-count-down 100))
        (cd2 (make-count-down 100)))
    (cd1 5)))
```